

# Control Statements

1. Decision Control Statements
2. Iterative Control Statements
3. Switch case Control Statements
4. Goto Control Statements.

# Decision Control Statements

- if
- if else
- Conditional Operator

```
int main()
{
    _____
    _____
    if ( )
    {
        _____
        _____
    }
    _____
    _____
    _____
}
```

if  
==

if (condition)

{

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

}

if  
block

T

F

यदि or अगर

यदि (TV)

{

पढ़ाई

}

Write a program to check whether a given number is positive or non positive.

5

positive

-3

Non-positive

0

Non Positive

① understand problem

② Test case

③ Dry run

## if-else

if ( condition )  
{

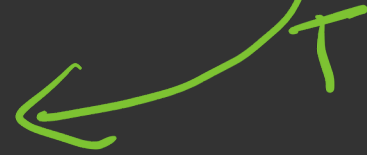
|||

}  
else

{

|||

}



F



यदि (TV)  
२

पढ़ाई

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नहीं तो  
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ज़हर

४

Write a program to check whether a given number is a positive or a non positive.

Ternary  
operator

Conditional Operator

expression 1 ? expression 2 : expression 3 ;



if (condition)

{

line 1

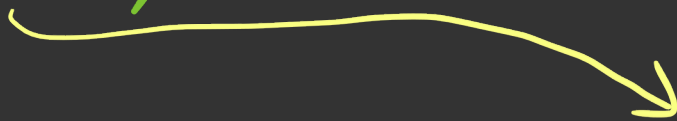
}

else

{

line 2

}



condition ? line 1 : line 2 ;

Write a program to check whether a given number is a positive or a non positive.

What is the difference between conditional operator and if else?  
conditional operator is an expression  
and if else is not an expression

```
x = if (-)
{
  -
}
else {
  -
}
```

$x = - ? - : - ;$

$x = a > b ? a : b ;$   
T  $\searrow$  F  $\searrow$   
 $x = a$   $x = b$

# Nested if else

```
if (condition)
{
    if (condition)
    {
        ==
    }
}
```

```
if (condition)
{
    if (condition)
    {
        ==
    }
    else
    {
        if (condition)
        {
            ==
        }
    }
}
```

Find output of the following program.

```
int main()
{
    int x = 5;
    if (x > 0)
    {
        printf("Hello");
    }
    printf("%d", x);
}
```

Find Output of the following program.

```
int main()
{
    int x=0;

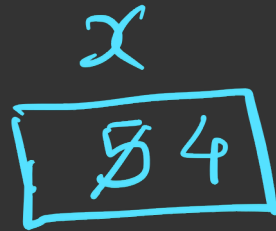
    if(x)
    {
        printf("Hello");

    }
    printf("%d", x);
}
```

Find Output of the following program.

```
int main()  
{
```

```
    int x=5;
```



```
    if (x=4)
```

```
        printf("Hello");
```

```
    else
```

```
        printf("Bye");
```

```
    printf("%d", x);
```

```
}
```

(a) Hello5

(b) Bye5

(c) Hello4

(d) Bye4

Find output of the following program.

```
int main()
{
    int x=5;
    if (x==4)
        printf("Hello");
    printf("%d", x);

    else
        printf("Bye");
    printf("%d", x);
}
```

if ( )

---

if ( )

{

}

==

else

{

}